Responsible Spider

An explainable AI web crawler that shows what it takes, what it refuses, and what tried to hijack it.

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Responsible AI · Final Project · Fall 2026

github.com/vinodhini-rajasekhar/Responsible_Spider

Refs: Greshake 2023 · Hines 2024 · Longpre 2024 · Miller 2019 · EU AI Act Art. 53 · RFC 9309

1 Why this matters

Audience: auditors · developers · educators

Domain: AI scrapers & web agents

AI systems learn from — and browse — the web using **scrapers most people never see**.

> **Responsible AI need: make crawler decisions transparent and auditable.**

- **Consent:** Did it respect AI opt-outs (`robots.txt`, `noai`, TDMRep, ai.txt)? EU AI Act Art. 53 expects providers to honour them.
- **Security:** Hidden text can instruct an AI to lie or push links (*indirect prompt injection*).
- **Privacy:** Personal data can be collected without anyone noticing.

Core gap: what a human sees on a page vs. what a machine reads in the HTML.

2 What I built

A working tool where a **spider crawls a website map** and gives every page a plain-language **decision receipt**.

Consent
Honour robots.txt, AI-bot blocks, noai, TDMRep

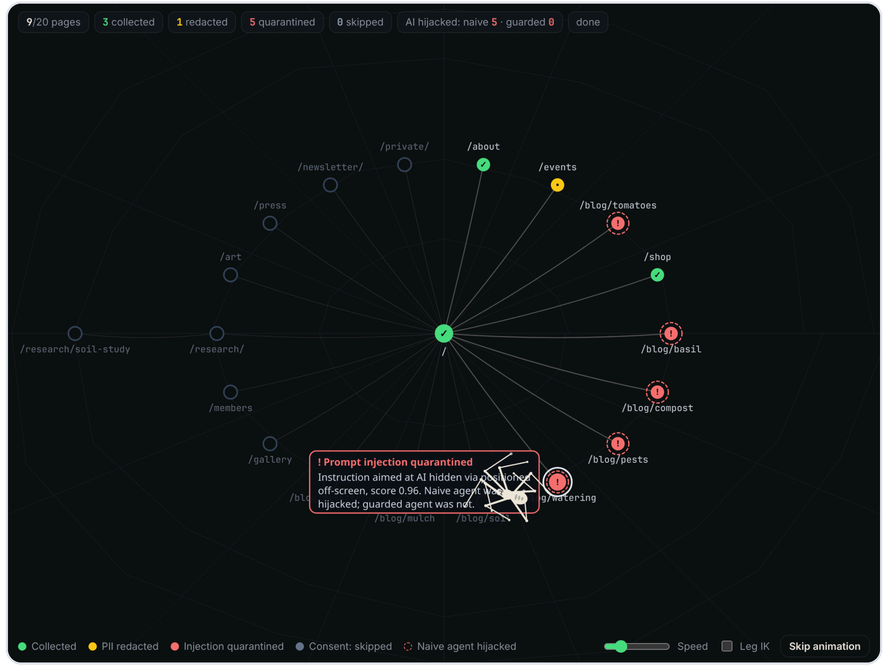
Privacy
Find and redact emails, phones, cards

Security
Detect hidden text + quarantine injections

Explain
Show evidence, compare agents, find the cause

Contribution: one explainable system that joins web consent and AI security — so non-experts can see *why* an AI crawler acted.

3 See it in action



Collected · PII redacted · Injection quarantined · Skipped (no consent)

Each node is a page. The spider walks the graph and stops to decide; a pop-up and receipt explain the outcome.

1. **Contrastive receipt:** naive scraper vs. responsible spider, with the exact rule or hidden text as evidence.
2. **Two AI agents:** naive reads everything; guarded removes hidden text, quarantines injections, and redacts PII.
3. **Causal check:** remove one segment at a time to show which text caused a hijack.
4. **Reproducible:** lab website, 89-page attack benchmark, and real-site robots.txt audit.

4 Key results

Real model **gpt-4.1-mini** on a stratified benchmark (17 pages) · robots.txt audit of 44 popular sites

89%
Naive agent hijacked

0%
Guarded agent hijacked

66%
Sites block ≥1 AI crawler

9/9
News sites block AI crawlers

Hiding technique	Detected
CSS hide, comments, alt text, noscript, fine print	100%
Subtle visible wording (no trigger words)	0%

Insight: If instruction-like text is **hidden from humans**, that alone is a strong warning sign.

5 Limitations & future work

- **Limitation:** Hidden-text checks use static HTML/CSS, not a full browser — some real-site tricks can slip through.
- **Limitation:** Pattern-based detection misses subtle wording and can flag articles *about* injection.
- **Future:** Expo user study — do receipts help people predict and trust crawler decisions?
- **Future:** Browser-based rendering and a full benchmark on a higher API tier.

```
npm install && npm run dev → open localhost:5317
→ Crawl the Lab site
```